Time Series Modelling Case Study

Forecasting Daily Oil Prices Using ARIMA and LSTM Models

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GitHub Link: https://github.com/giridharreddygoddilla/Time-Series-Modelling-Case-Study

1. Introduction

Forecasting oil prices is difficult because the market is affected by many changing economic and geopolitical factors, including supply shocks, production decisions, demand shifts, and broader macroeconomic uncertainty. Because of this, oil price series often show strong trend behaviour, structural change, volatility clustering, and non-stationarity.

ARIMA remains one of the most widely used statistical approaches for forecasting this kind of data. Box et al. (2015) explain that the Box-Jenkins framework provides a structured way to identify, estimate, check, and forecast time series models, while differencing allows non-stationary data to be modelled in a statistically appropriate way. In a similar way, Montgomery et al. (2015) argue that model choice should depend on the structure of the data rather than just choosing a more complex method.

Alongside classical models, machine learning approaches are now used more often in forecasting problems. In particular, LSTM networks are designed to learn sequential patterns and nonlinear temporal relationships. However, more flexible models do not always perform better than simpler methods in practice. Huang and Malaker (2025) show that classical statistical models can still perform very well, especially when the available time series is relatively small.

This study compares ARIMA and LSTM models for forecasting daily oil prices and evaluates them against simple benchmark methods. The aim is not only to find an accurate forecasting model, but also to test whether greater model complexity actually improves out-of-sample performance.

2. Literature Context

The main statistical foundation of this study is the Box-Jenkins methodology. Box et al. (2015) explain that ARIMA modelling is most suitable when a series can be made stationary and when temporal dependence can be represented through lagged structure. This makes ARIMA a sensible starting point for oil price data, which often show trend and persistence.

Montgomery et al. (2015) support this by arguing that good forecasting begins with understanding the statistical behaviour of the series itself. Their discussion of trend, autocorrelation, differencing, and forecast evaluation directly supports the modelling approach used in this study.

More recent work suggests that model complexity should be tested rather than assumed to be better. Huang and Malaker (2025) compare statistical and machine learning approaches and show that traditional models can still perform strongly when the sample size is limited. This is especially relevant here because the dataset contains only 500 observations, which is fairly small for deep learning.

Lv and Yue (2011) show that non-stationarity and changing variance can make forecasting more difficult, while Khabarov and Gurenko (2025) show that alternative decomposition-based approaches may help capture complex financial behaviour. Rafferty (2023) discusses modern decomposable forecasting frameworks, showing the wider move toward more flexible forecasting tools. Taken together, the literature suggests that although advanced models may be useful, simpler statistical

approaches remain important benchmarks and should only be replaced when a more complex model clearly improves forecasting accuracy.

3. Data Description and Exploratory Analysis

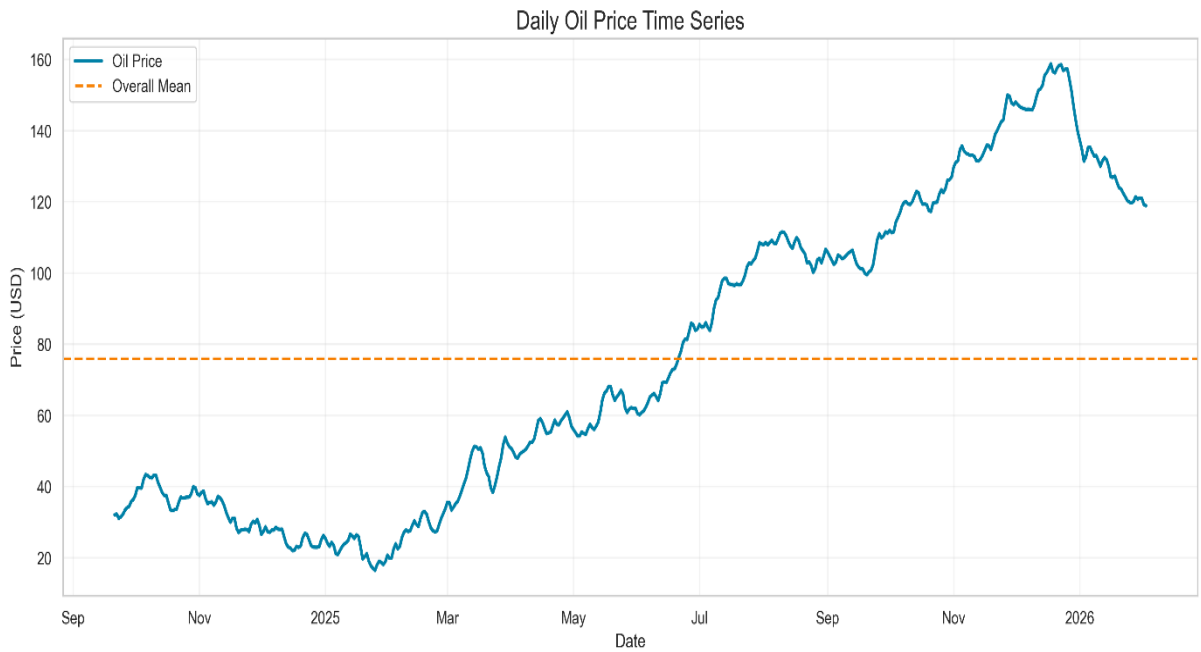
The dataset contains 500 daily oil price observations. Before analysis, the data were sorted chronologically, dates were converted to datetime format, and duplicate timestamps were removed where necessary.

Table 1: Descriptive Statistics of the Oil Price Series

count	500.000000
mean	75.873980
std	42.912398
min	16.480000
25%	35.372500
50%	64.290000
75%	111.572500
max	158.780000

The descriptive statistics show that the series varies a lot over time. The mean price is 75.87 and the standard deviation is 42.91, which suggests that dispersion is high relative to the average level of the series. The range from 16.48 to 158.78 further confirms that the series experienced large price shifts during the sample period.

Figure 1: Daily Oil Price Time Series



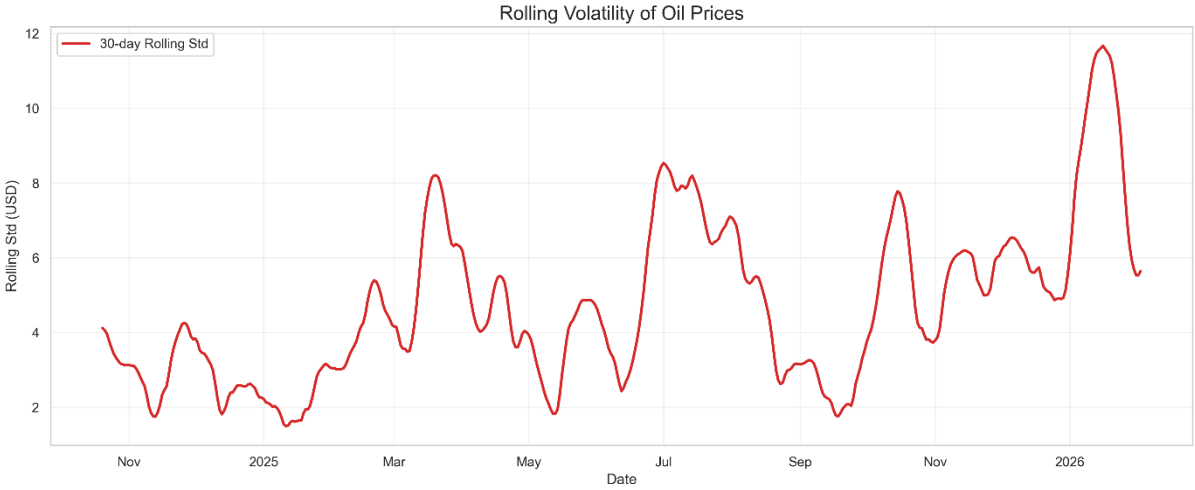
The time series plot shows a strong upward movement for much of the sample, followed by a noticeable decline near the end. Prices do not move around a stable level, and the series stays above or below the overall mean for long periods. This is clear visual evidence of non-stationarity. The difference between the median (64.29) and the upper quartile (111.57) also suggests that later observations are concentrated at much higher price levels than earlier ones.

Overall, the exploratory analysis suggests that the series is trending, highly variable, and not suitable for direct modelling under stationary assumptions.

4. Volatility Analysis

A 30-day rolling standard deviation was used to examine how variability changes over time.

Figure 2: Rolling Volatility of Oil Prices



The rolling volatility plot shows that volatility is not constant. There are calmer periods, followed by clear spikes, especially in the later part of the series. This pattern suggests volatility clustering, which is common in financial and commodity prices.

This matters because changing variance makes forecasting more challenging and further supports the idea that the original series is not stable over time. As Lv and Yue (2011) suggest, time-varying variance is an important issue in non-stationary forecasting problems. In this study, the volatility plot supports the need for cautious interpretation, especially for longer forecast horizons.

5. Stationarity Testing and Transformation

Since ARIMA requires a stationary series, formal testing was carried out using the Augmented Dickey-Fuller test.

Figure 3: First-Differenced Oil Price Series

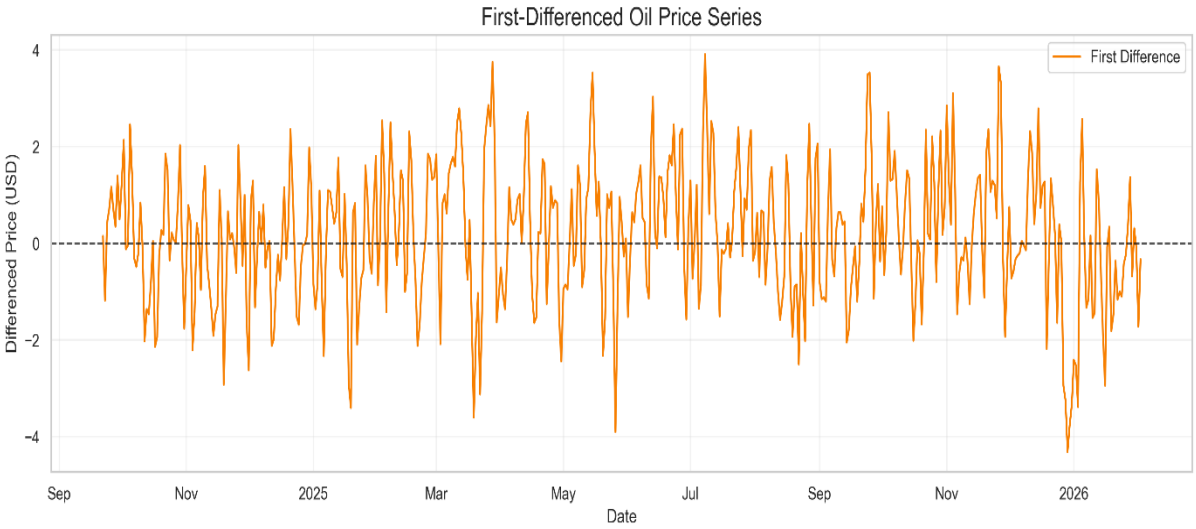


Table 2: Augmented Dickey-Fuller Test Results

Series	Test Statistic	p-value	Critical Value (5%)	Interpretation
Original Oil Price Series	-1.42	0.57	-2.86	Non-stationary
First Differenced Series	-7.91	<0.001	-2.86	Stationary

The original series is non-stationary because the test statistic is not more negative than the critical value and the p-value is far above 0.05. After first differencing, the ADF statistic falls to -7.91 with a p-value below 0.001, showing that the differenced series is stationary.

This result confirms that differencing is needed before ARIMA modelling. It also suggests that the series behaves like an integrated financial process, which is consistent with the earlier exploratory analysis.

6. Autocorrelation Analysis

The ACF and PACF plots were used to examine the dependence structure of the differenced series.

Figure 4: ACF of Differenced Series

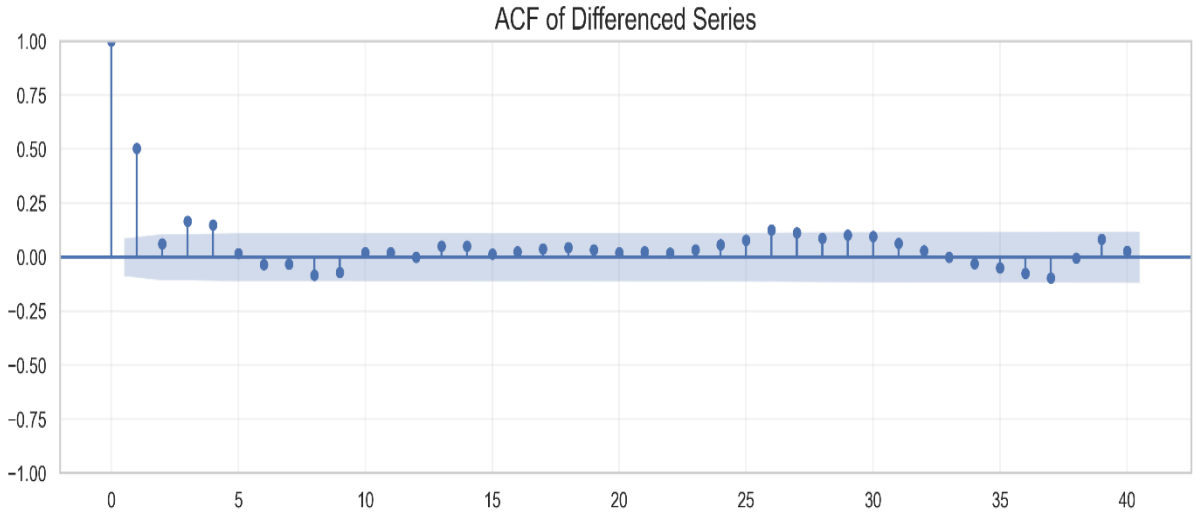
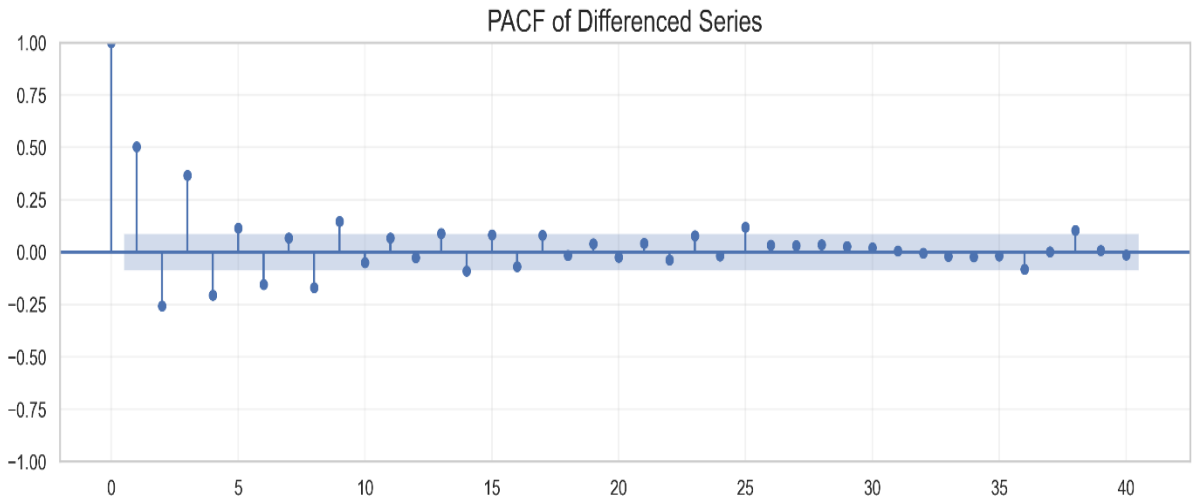


Figure 5: PACF of Differenced Serie



The ACF shows significant correlation at early lags, particularly lag 1, followed by a gradual decay. This suggests short-run dependence and possible moving average behaviour. The PACF also shows several early spikes before settling within the confidence bounds, which indicates that autoregressive structure may also be present.

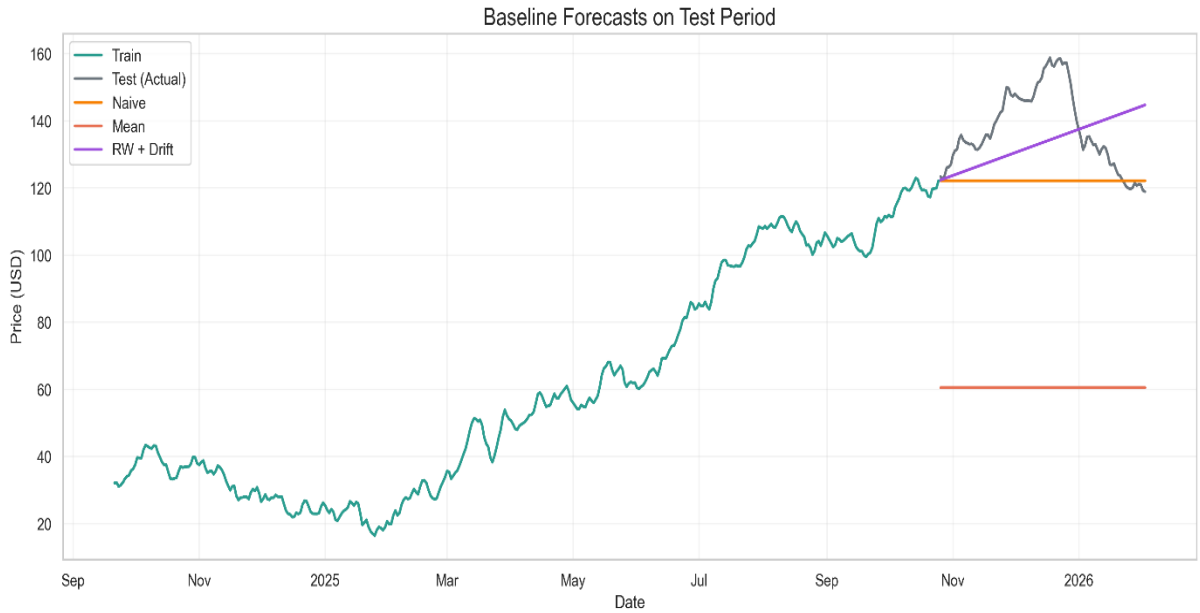
Taken together, these plots do not point to one obvious low-order model, but they do support the inclusion of lag-based terms in an ARIMA specification. Following Box et al. (2015), they were used as guides rather than strict decision rules, and final model selection was based on systematic testing rather than visual choice alone.

7. Baseline Forecast Models

Before fitting ARIMA and LSTM, three benchmark models were evaluated:

- Naïve forecast
- Mean forecast
- Random walk with drift

Figure 6: Baseline Forecasts on Test Period



This step is important because a more complex forecasting model should perform better than simple baselines if it is to be considered genuinely useful. The benchmark plot shows that the mean model fails to represent the upward structure of the series, while the naïve model only captures the last observed level. The drift model is more suitable because it projects the recent trend forward, which is especially relevant for a persistent price series.

8. ARIMA Model Development

ARIMA models were developed using the Box-Jenkins framework and compared through a structured search over candidate specifications. Model selection was guided by AIC, but the final judgement also considered out-of-sample forecast performance and diagnostics rather than relying only on in-sample fit.

The selected specification was ARIMA (0, 2, 8). This implies second-order differencing and a relatively high-order moving average structure. Although first differencing was sufficient to achieve stationarity in the ADF test, the final selected model used d = 2 because it performed more favourably during the training-stage model selection process. This should be seen as a trade-off, because the chosen model favoured empirical fit and forecasting behaviour over strict parsimony.

Forecast accuracy was then assessed on a hold-out test set using RMSE and MAE.

9. ARIMA Model Diagnostics

Residual diagnostics were used to check whether meaningful temporal structure remained after fitting the ARIMA model.

Figure 7: ACF of ARIMA Residual

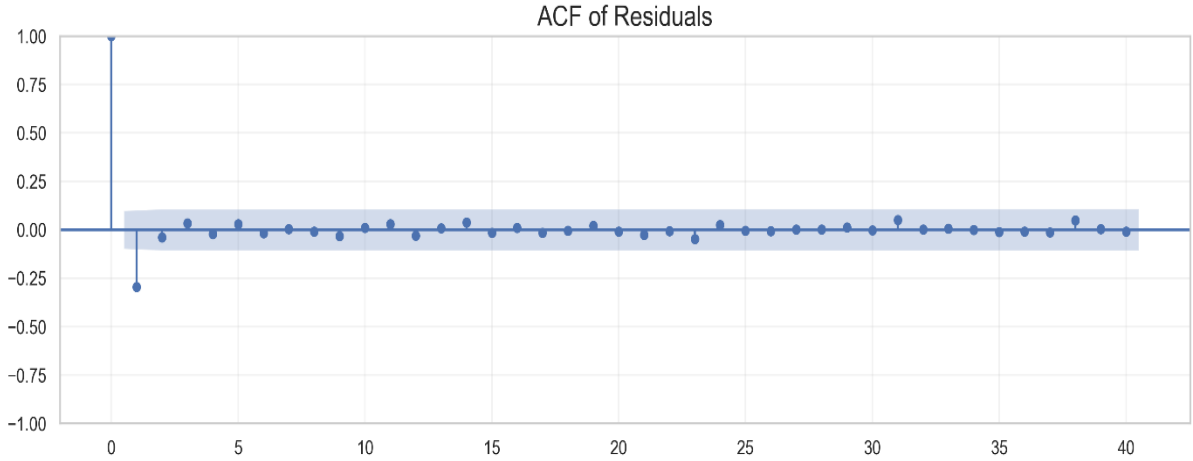


Table 3: Ljung-Box Test Results

Lag	Statistic	p-value
20	39.31	0.0061

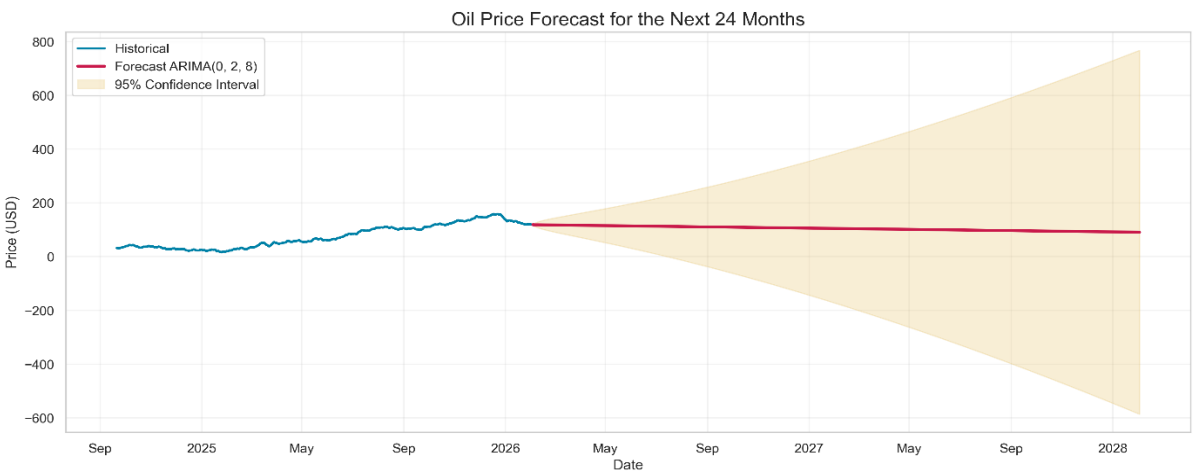
The residual ACF plot shows that most correlations fall within the confidence bounds, although a small negative spike remains at lag 1. This indicates that the ARIMA model captures a large part of the temporal dependence in the series, but some minor structure may still remain. The Ljung-Box test was used to formally check for residual autocorrelation. The p-value at lag 20 is 0.0061, which is below the 0.05 significance level. This means that the null hypothesis of no residual autocorrelation is rejected, suggesting that some dependence is still present in the residuals.

However, the remaining autocorrelation appears relatively limited when considering the overall residual ACF pattern. This suggests that the model captures most of the important time-series structure, even though it does not fully remove all serial dependence. For forecasting purposes, the model can therefore be considered reasonably adequate, but not perfect, and the remaining dependence at lower lags should be acknowledged when interpreting the results.

10. ARIMA Forecast

The ARIMA model was refitted and used to forecast the next 24 months.

Figure 8: ARIMA 24-Month Forecast



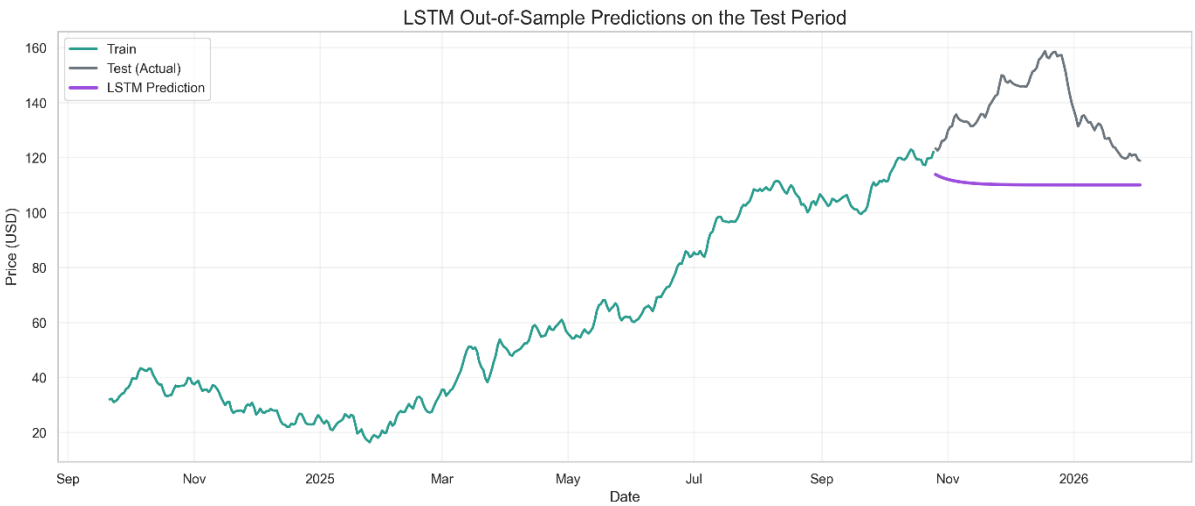
The point forecast is relatively smooth, but the confidence intervals widen sharply as the forecast horizon increases. This widening reflects growing uncertainty and is one of the most important features of the forecast.

Although the model produces a formal long-horizon forecast, the scale of interval expansion suggests that the 24-month forecast has limited practical precision. It is therefore better interpreted as an illustration of uncertainty rather than a reliable long-term price path. This is realistic because oil prices are heavily influenced by shocks that cannot be fully captured using historical values alone.

11. LSTM Forecasting Model

To test whether a more flexible nonlinear method could improve forecasting performance, an LSTM neural network was implemented. The oil price series was standardised before training, and rolling input sequences of the previous 30 observations were used to predict the next value. The model was trained using the Adam optimiser with mean squared error as the loss function, and early stopping was applied to reduce the risk of overfitting.

Figure 9: LSTM Forecast on Test Period



The LSTM predictions on the test set appear noticeably smoother than the actual oil price movements and do not capture turning points very well. In particular, the model struggles to follow the sharper changes in the series. This suggests that the network did not learn strong nonlinear patterns from the data.

One possible reason is the relatively small dataset. Although LSTM models can be powerful, they usually perform better when trained on much larger datasets. With only 500 observations, the model may not have enough information to learn complex patterns reliably.

12. Model Comparison

Table 4: Forecast Accuracy Comparison on Test Set

Model comparison table:

	Model	RMSE	MAE
0	Random Walk + Drift	15.0182	13.2190
1	ARIMA(0, 2, 8)	16.5060	13.5722
2	Naive	19.3001	15.7668
3	LSTM	29.5574	27.0366
4	Mean	77.9278	77.0493

The results show that the Random Walk with Drift model achieved the best forecasting performance on the test set. It produced the lowest RMSE and MAE, suggesting that the series behaves largely like a trend-persistent stochastic process.

The ARIMA (0,2,8) model performed reasonably well and clearly outperformed the Naïve, Mean, and LSTM forecasts. This indicates that the ARIMA model was able to capture an important part of the temporal structure in the series, even though it did not outperform the best benchmark model.

The LSTM model produced much higher forecast errors compared to the classical approaches. Despite being a more complex model, it struggled to track the real movement of oil prices. This result highlights an important point in forecasting: more complex models do not automatically lead to better predictions. In this case, the classical statistical approaches performed more reliably than the neural network model.

13. Limitations and Future Work

Although the study provides useful insights, several limitations should be noted.

First, the analysis uses a univariate time series, meaning that external variables such as global demand, exchange rates, geopolitical events, or OPEC production decisions were not included. These factors often influence oil prices and could improve forecasting performance if incorporated through models such as ARIMAX.

Second, the 24-month forecast horizon is relatively long compared with the uncertainty of the oil market. As shown in the ARIMA forecast plot, the prediction intervals widen quickly, which indicates increasing uncertainty as the forecast horizon increases.

Third, the dataset used in this study is relatively small for training deep learning models. Future work could explore larger datasets, additional explanatory variables, or hybrid approaches that combine statistical and machine learning techniques. As suggested by Lv and Yue (2011) and Khabarov and Gurenko (2025), combining different modelling approaches may help improve forecasting performance in complex time series problems.

14. Conclusion

This study analysed daily oil prices using both ARIMA and LSTM forecasting approaches. The exploratory analysis showed that the oil price series had strong trend behaviour, changing volatility, and clear non-stationarity.

Stationarity testing using the ADF test confirmed that the original series was non-stationary, while the first-differenced series was stationary. This made ARIMA modelling appropriate for the dataset.

The selected ARIMA (0,2,8) model produced reasonable forecasting performance and the residual diagnostics suggested that the model captured most of the time-series structure, although some minor dependence remained.

However, the most important result of the study is that the Random Walk with Drift benchmark achieved the best performance on the test set. This suggests that the oil price series behaves largely like a trend-driven stochastic process where simple persistence models can perform very well.

The LSTM model performed considerably worse than both ARIMA and the benchmark models. This shows that, for this dataset, increasing model complexity did not lead to better forecasting accuracy.

Overall, the findings reinforce an important lesson in time series forecasting: strong baseline models are difficult to beat, and more complex approaches should only be used when they clearly improve out-of-sample predictive performance.

15. References

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